probability distributions

→ When analysing distributions,it is important to understand what kind of data we have

population data

↓

referring to all The Data

sample data

↓

just a part of it

→ a constant relationship exists between mean and variance for any distribution

$$\sigma^2 = E((Y-\mu^2)) = E(Y^2) - \mu^2$$

Types of probability distributions

discrete distributions

--picking a card

→continuous distributions

-distance
-Time

Discrete distributions

uniform
Bernoulli
Binomial
poisson

continuous distributions

Normal
student's T
chi squared
Exponential
logistic

What is a Uniform Distribution?

A uniform distribution, also called a rectangular distribution, is a probability distribution that has constant probability

The distribution is defined by two parameters:

minimum, a.
maximum, b.

The expected value (or mean) of a uniform random variable X is:

$E(X) = (1/2)(a + b)$

The variance of a uniform random variable is:

$Var(x) = (1/12)(b-a)^2$

What is a Bernoulli Distribution?

A Bernoulli distribution is a discrete probability distribution for a Bernoulli trial — a random experiment that has only two outcomes (usually called a "Success" or a "Failure"). For example, the probability of getting a heads (a "success") while flipping a coin is 0.5. The probability of "failure" is 1 – P (1 minus the probability of success, which also equals 0.5 for a coin toss).

It is a special case of the binomial distribution for n = 1. In other words, it is a binomial distribution with a single trial

$E[X]= P$

$Var[X]=P(1-P)$

The binomial distribution evaluates the probability for an outcome to either succeed or fail

which means you either have one or the other — but not both at the same time. For example, you either win the lottery or you don't

The binomial distribution formula is:
b(x; n, P) = nCx * P^x * (1 − P)^n − x

A Poisson distribution is a tool that helps to predict the probability of certain events happening when you know how often the event has occurred

Calculating the Poisson Distribution The Poisson Distribution pmf is:

$P(x; \mu) = (e^{-\mu} * \mu^{x}) / x!$

μ (the expected number of occurrences) is sometimes written as λ

The normal distribution

is a probability distribution that is symmetric about its center

Properties of a normal distribution

The mean, mode and median are all equal.

The curve is symmetric at the center (i.e. around the mean, μ).

Exactly half of the values are to the left of center and exactly half the values are to the right.

The total area under the curve is 1, or 100%. In other words, the curve represents 100% of all possible data.

student's T distribution

t(k)

small sample size approximation of
a normal bistribution

mean
variance
degrees of
freedom → k

E(Y)= M
Var(Y)= S^2.k/k-2

frequently used when conducting statistical analysis

Hypothesis testing with limited data

CDF table

Chi-squared distribution

$Y \sim t(k)$

Asymmetric

$E(X)=K$

$Var(X)=2K$

Conditional probability

What happens next depends on what already happened

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Multiplication rule

$$P(E_1)\, P(E_2|E_1) \ldots P(E_n)$$

$$= P(E_1 E_2 \ldots E_n)$$

Bayes's Formula

$$P(A/B) = \frac{P(B/A)\, P(A)}{P(B)}$$

Odds

$$\frac{P(A)}{P(A^c)}$$

Mutually exclusive

$$P(A \cap B) = 0$$

Independent

$$P(A \cap B) = P(A) \cdot P(B)$$

$$P(AB) = P(A)P(B)$$

$$P(A/B) = P(A)$$

$$P(B/A) = P(B)$$